# Introduction to Illumina sequencing and mNGS data analysis

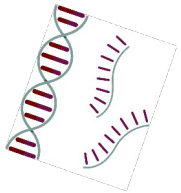
# mNGS pipeline

## Study design



- Type(s)
- Controls
- Storage
- Lab setup

## xNA extraction



- Type(s)
- Controls
- QC

## Library prep



- Host removal
- Fragmentation
- dsDNA generation
- End-repair
- Adapter ligation
- Barcoding & PCR
- Clean-up
- Pooling
- QC

## Sequencing



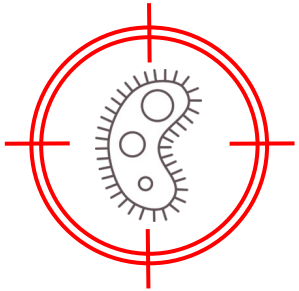
- Machine
- Read length
- Loading conc.

## Data analysis



- QC data
- Host removal
- Alignment
- Blast
- Ranking of hits

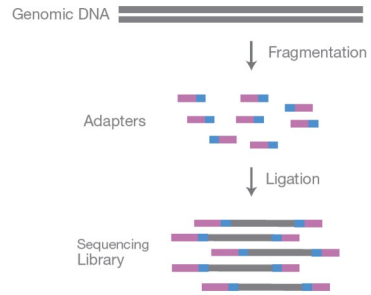
## Pathogen ID



**INTRODUCTION**

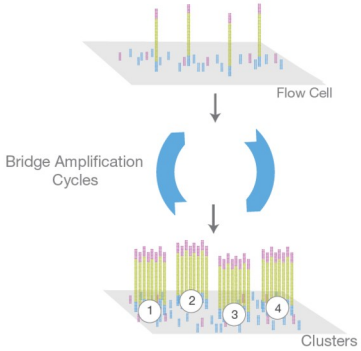
# Introduction to illumina® Next-Generation sequencing (NGS)

## A. Library Preparation



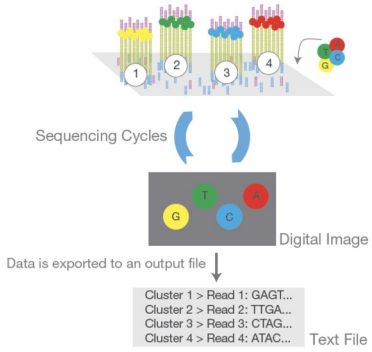

NGS library is prepared by fragmenting a gDNA sample and ligating specialized adapters to both fragment ends.

## B. Cluster Amplification



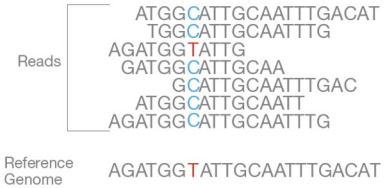

Library is loaded into a flow cell and the fragments hybridize to the flow cell surface. Each bound fragment is amplified into a clonal cluster through bridge amplification.

## C. Sequencing




Sequencing reagents, including fluorescently labeled nucleotides, are added and the first base is incorporated. The flow cell is imaged and the emission from each cluster is recorded. The emission wavelength and intensity are used to identify the base. This cycle is repeated "n" times to create a read length of "n" bases.

## D. Alignment & Data Anaylsis



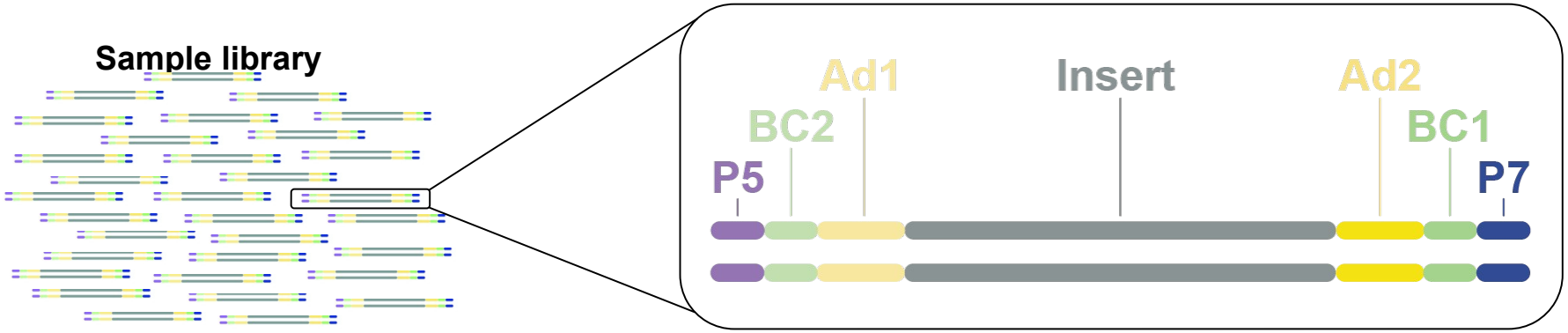

Reads are aligned to a reference sequence with bioinformatics software. After alignment, differences between the reference genome and the newly sequenced reads can be identified.

# illumina® Next-generation sequencing (NGS)

- What is a library?
- What is a read?
- What is paired-end sequencing?
- What is a cluster?
- What is (de-)multiplexing?
- What is a fastq file?
- What is coverage?

# Sequencing library = collection of fragments ready for sequencing

1. DNA is fragmented into smaller pieces, optimal for sequencing
2. Adapter sequences, barcode sequences and flow cell binding sites are added to allow the sequencing process to occur




**Insert** = DNA fragment to be sequenced

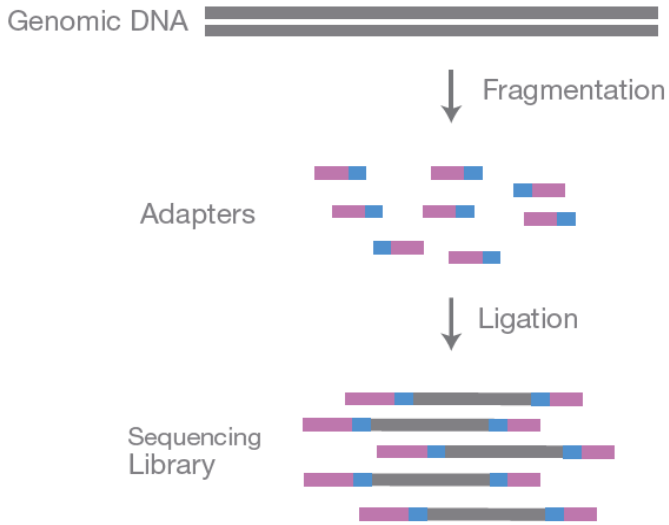
**Ad1/Ad2** = Adapters that allow initiation of sequencing

**BC1/BC2** = Barcodes specific for each sample (one for each strand)

**P5/P7** = Flow cell binding sites

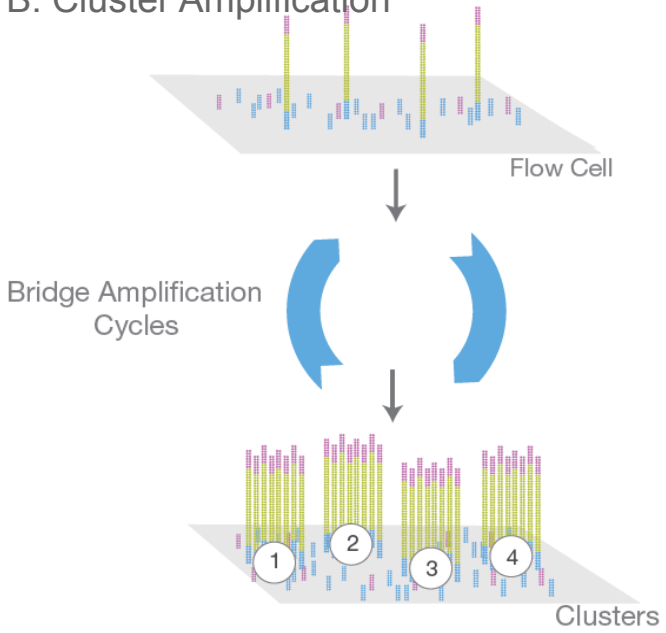
# Cluster formation through bridge amplification

## A. Library Preparation




NGS library is prepared by fragmenting a gDNA sample and ligating specialized adapters to both fragment ends.

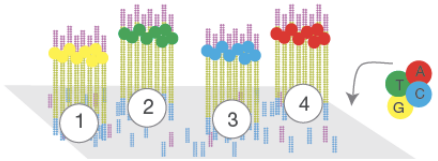
## B. Cluster Amplification




Library is loaded into a flow cell and the fragments hybridize to the flow cell surface. Each bound fragment is amplified into a clonal cluster through bridge amplification.

# illumina® Fluorescence imaging

## C. Sequencing




Sequencing Cycles



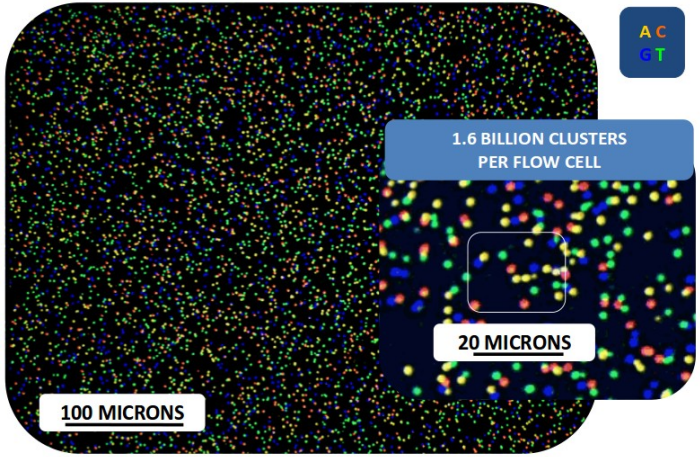

Digital Image

Data is exported to an output file

Cluster 1 > Read 1: GAGT...
Cluster 2 > Read 2: TTGA...
Cluster 3 > Read 3: CTAG...
Cluster 4 > Read 4: ATAC...

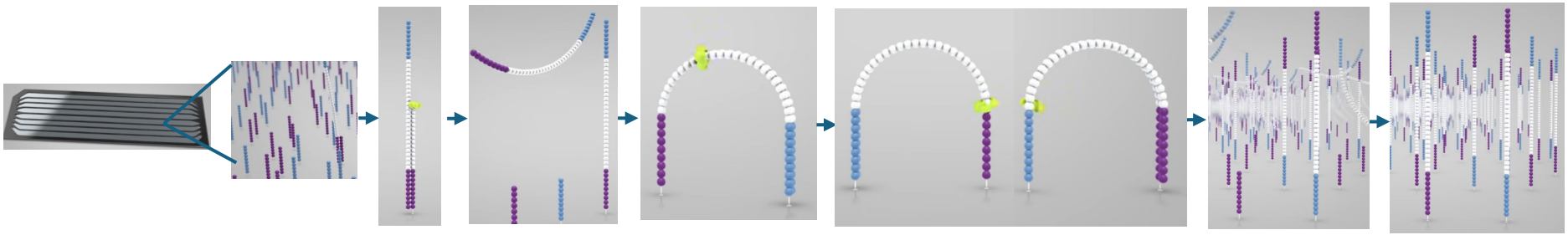
Text File

Sequencing reagents, including fluorescently labeled nucleotides, are added and the first base is incorporated. The flow cell is imaged and the emission from each cluster is recorded. The emission wavelength and intensity are used to identify the base. This cycle is repeated “n” times to create a read length of “n” bases.

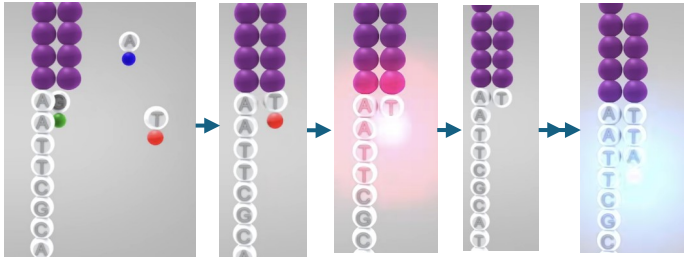


# illumina® Sequencing By Synthesis (SBS)

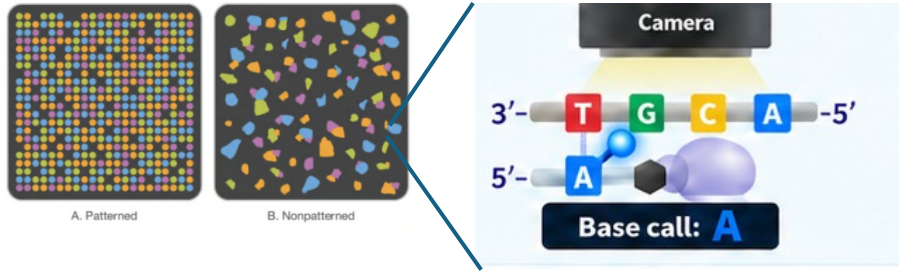
1.Cluster formation through bridge amplification: the result is a lawn of clonal ssDNA ‘islands’



2.Nucleotides emit in each incorporation cycle

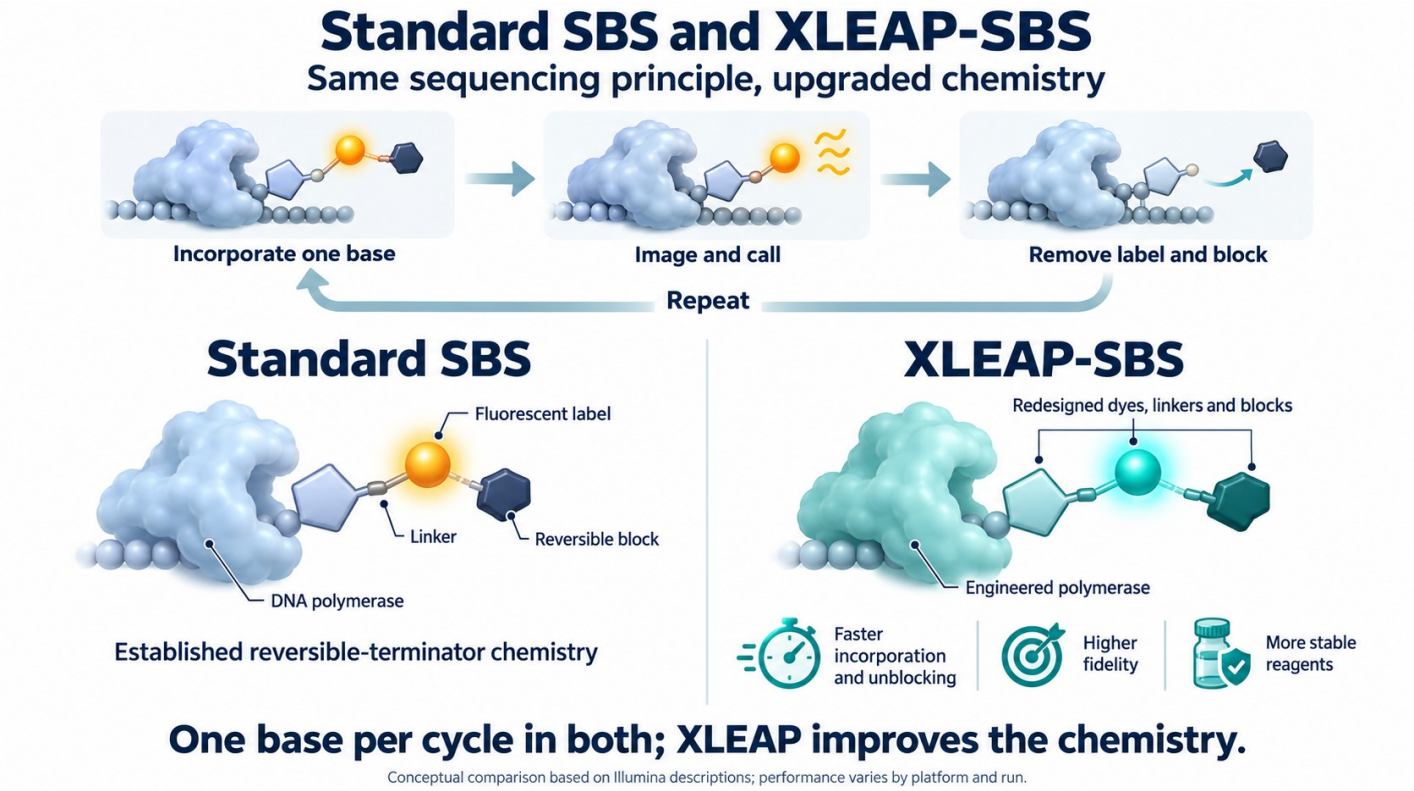


3.Sequencing is done by imaging fluorescence at each cycle

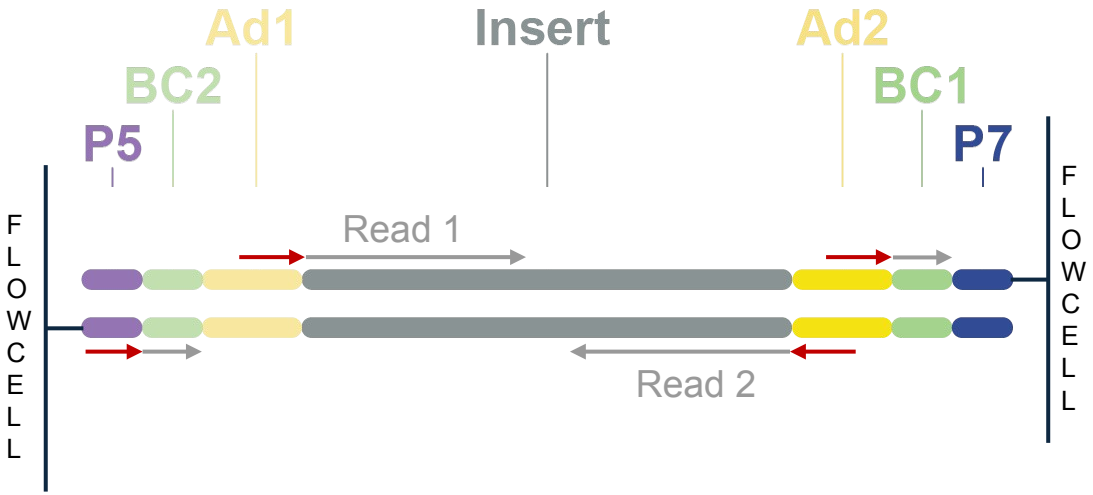





# illumina® XLEAP-SBS Chemistry update



# Sequencing read order



# illumina® Paired-End sequencing

Paired-end sequencing

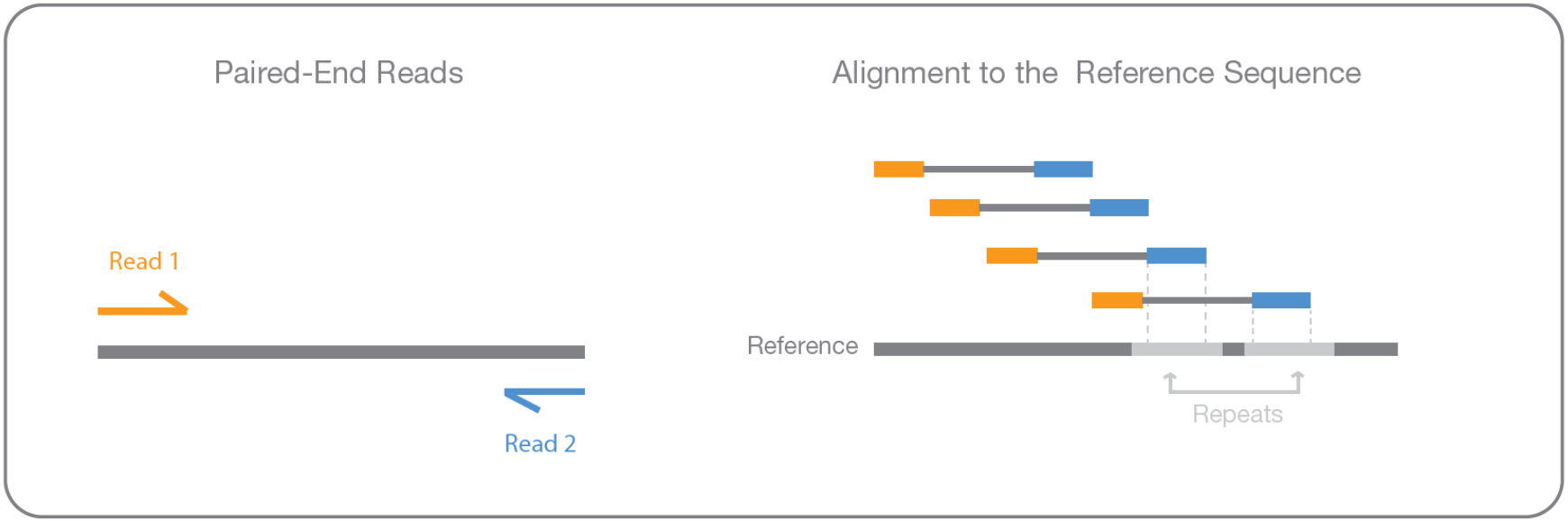



**Figure 4: Paired-End Sequencing and Alignment**—Paired-end sequencing enables both ends of the DNA fragment to be sequenced. Because the distance between each paired read is known, alignment algorithms can use this information to map the reads over repetitive regions more precisely. This results in much better alignment of the reads, especially across difficult-to-sequence, repetitive regions of the genome.

# illumina® Pooling and demultiplexing

Multiplexing or Pooling = Sequencing multiple samples together in one reaction

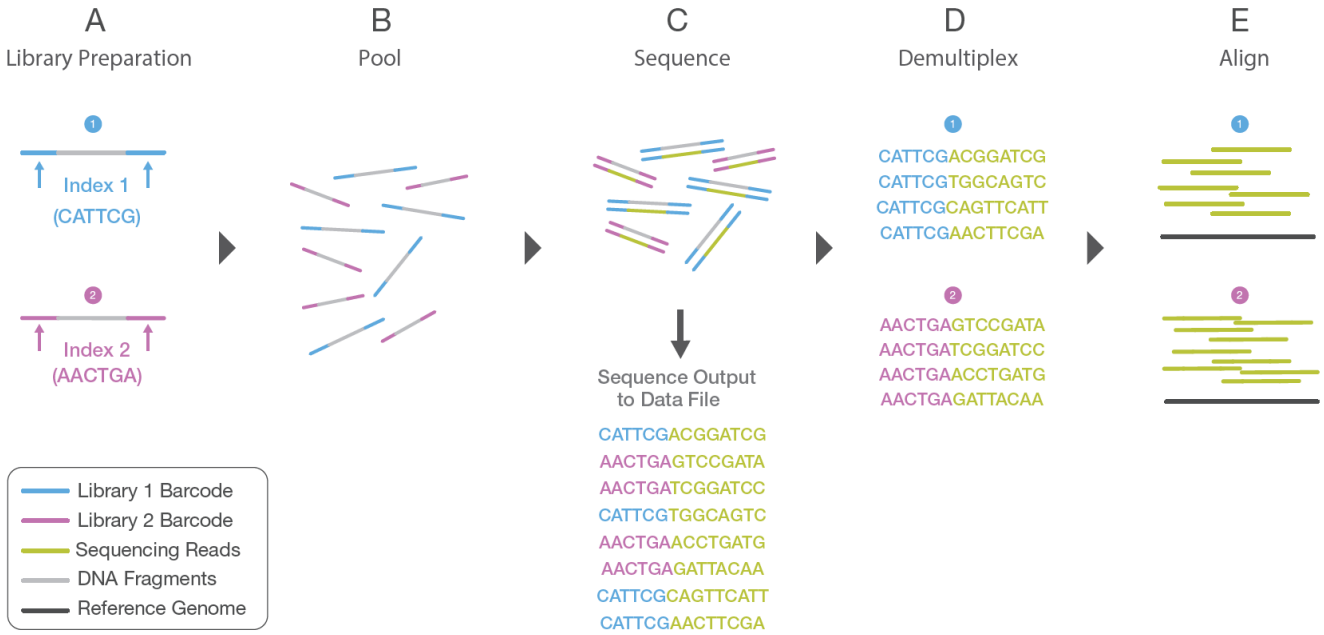



**Figure 5: Library Multiplexing Overview.**

a. Two distinct libraries are attached to unique index sequences. Index sequences are attached during library preparation.

b. Libraries are pooled together and loaded into the same flow cell lane.

c. Libraries are sequenced together during a single instrument run. All sequences are exported to a single output file.

d. A demultiplexing algorithm sorts the reads into different files according to their indexes.

e. Each set of reads is aligned to the appropriate reference sequence.

# fastq file = raw data file containing sequencing data, generated by sequencer

Sequencer will demultiplex sequencing data, resulting in a two fastq files per sample (read 1 and read 2)

```
READ 1 .FASTQ FILES

@NB501961:233:H2VKMBGXB:1:11101:16944:1057 1:N:0:GAGTTCCGACAT+NGTGGAGGATTG
CGGCTNTTCCGTCGGTGTACGCGGTTTTGAGTCGNNN
```

# fastq file = raw data file containing sequencing data, generated by sequencer

Example Read1.fastq

```
@NB501961:14:HM7TLBGX2:1:11102:3233:17234 1:N:0:GATCACCA+GATCACCA
CATTCGGCTGGGTTTCGTCACCCTGCGGGAAGATGCGGGTCCAGGCGATAGAGGTGCGGAAGCATTTGAAGCCCA
+
AAAAAEEEEEEEEEE/EEEEEEEEEEEEEEEEEEEEE<EEEEEEEEE//EAEEEEEEAA/EEE/EEAAEEE<6/
```

Each read is described by 4 elements in 4 lines:

1. A sequence identifier with information about the run (always starts with '@')

```
@<instrument>:<run number>:<flowcell ID>:
<lane>:<tile>:<x-pos>:<y-pos> <read>:<is
filtered>:<control number>:<sample number>
```

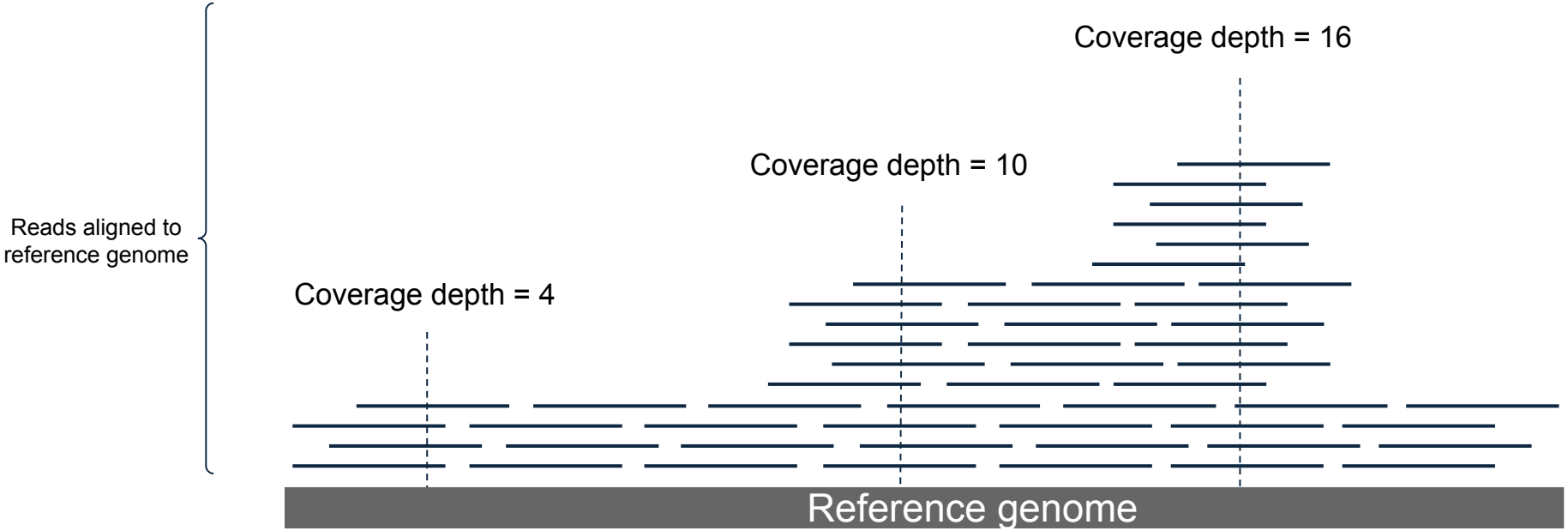
1. The base calls: A, C, T, G and N
2. A separator, which is simply a plus (+) sign
3. The base call quality scores (coded in ascii format)

| Quality Score | Probability of Incorrect Base Call | Inferred Base Call Accuracy |
|---|---|---|
| 10 (Q10) | 1 in 10 | 90% |
| 20 (Q20) | 1 in 100 | 99% |
| 30 (Q30) | 1 in 1000 | 99.90% |

$Q = -10\log_{10}(e)$

# Coverage = How many times a base has been sequenced

Coverage = How many times each base has been sequenced on average throughout the genome



Next: 07-I_ iSeq 100 / MiSeq loading and sequencing